Subject Description Form

Subject Code	**ITC4084E**
Subject Title	**Smart Textiles for Wearable Electronics**
Credit Value	**3**
Level	**4**
Pre-requisite/ <Co-requisite> / (Exclusion)	Nil
Objectives	The subject aims to provide students with the knowledge of the latest development (materials, technologies and products) in smart textiles for wearable technologies. The subject is designed to introduce the concepts, principles, technological processes for the development of new smart textile and wearable products, which enable students to identify the features and requirements of smart textiles and wearable products. The subject also provides an opportunity to apply the skills and knowledge to analyse the broad range of latest developments critically from wearable product design perspectives.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) assess technologies involved in smart textile and wearable electronic products and manufacturing processes, their advantages and disadvantages; (b) describe the development trends in technologies for smart textile and wearable electronics and their potentials in industrial applications. (c) cooperate efficiently in a team to achieve goals; (d) develop critical and creative thinking; (e) develop the ability to adapt new technologies in wearable product design and update her/his knowledge; (f) develop the ability for sourcing, selecting and integrating information.
Subject Synopsis/ Indicative Syllabus	**(I)** **<u>Smart Fibers and Fibrous Assembles</u>** Electric and photonic functions: electrically conductive fibers, optical fibers; photochromic fibers; electrochromic fibers; actuation fibers. **(II)** **<u>Interactive Textile Devices</u>** Fiber-based electronic circuit and systems; flexible fabric strain sensors; textile electrodes for bio-potentials; textile energy harvesters; textile energy storages; fabric displays; morphing textile fabrics; textile circuit boards.

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<tr><td></td><td>(III) <u>Wearable Products Design and System Analysis</u>
Thermal regulating garment; health monitoring garment; i-shoes; wearable phototherapy device.</td></tr>
<tr><td>Teaching/Learning Methodology</td><td>Lectures will be used to introduce the principles, applications and decision implications of various technological developments to students.

Laboratory/studio-based case-studies will be used to provide hand-on experience and supplement the lecture program.

Students will be encouraged to interact with the lecturer and present their own investigations of the new developments and products. Individual laboratory/studio projects and group projects will be the major parts of continuous assessment components.</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>1. Coursework</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td></td><td>✓</td><td></td><td></td></tr>
<tr><td>Total</td><td>100%</td><td colspan="6"></td></tr>
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In the continuous assessment, by group projects, we focus on all rounded education. That is, in addition to professional knowledge, we also stress the importance of teamwork, critical and creative thinking, the ability to adapt new technologies and update her/his knowledge as well as the ability for sourcing, selecting and integrating information. For example, interaction among group members is a very good way for enhancing their communication skills, practice their team work spirit, leadership as well as entrepreneurship. In order to gain systematic technical knowledge/skills, students are also given some questions to answer as assignments.

In the examination, we assess students’ learning performance in terms of defining concepts, explaining principles and describing and analysing application examples as well as providing rationales for smart textile and wearable products and integration processes.</td></tr>
<tr><td rowspan="7">Student Study Effort Expected</td><td>Class contact:</td><td></td></tr>
<tr><td>▪ Lecture</td><td>12 Hrs.</td></tr>
<tr><td>▪ Studio/Laboratory</td><td>26 Hrs.</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>▪ Projects</td><td>25 Hrs.</td></tr>
<tr><td>▪ Self study</td><td>42 Hrs.</td></tr>
<tr><td>Total student study effort</td><td>105 Hrs.</td></tr>
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Reading List and References	**Recommended Textbook** **Essential** Xiaoming Tao, Handbook of smart textiles, Springer, 2015. (e-version available) Abdelsalam (Sumi) Helal, Mounir Mokhtari, Bessam Abdulrazak, The Engineering Handbook of Smart Technology for Aging, Disability, and Independence, John Wiley & Sons, Inc. 2008. (e-version available) **Supplementary** Xiaomong Tao, Smart fibres, fabrics and clothing – fundamentals and applications, Elsevier, 2001. **Periodicals and Websites** Textile Research Journal Smart Materials Bulletin Smart Materials and Structures High Performance Textiles Textile Outlook International Medical textiles